

Kamsi Nwabueze

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EDUCATION

University of Southern California

Los Angeles, CA

B.S. in Computer Engineering and Computer Science

May 2028

- Relevant Coursework: Data Structures, Parallel and Distributed Computation, Distributed Systems and IoT, Linear Algebra, Professional C++, Functional Programming, Theory of Computation and Algorithms

TECHNICAL SKILLS

Languages & Frameworks: Go, Python, C++/ C, JavaScript, TypeScript, Java, Node.js, Flask, React

Tools & Platforms: Git, Docker, Linux, Bash, AWS, PyTorch, TensorFlow, Scikit-learn

WORK EXPERIENCE

Vitel Wireless Limited

Los Angeles, CA

Software Engineering Intern

June 2025 – August 2025

- Designed and built Veye, a modular ONVIF-compatible home security platform integrating 4G IP cameras with cloud processing for large-scale remote monitoring
- Integrated the Shinobi open-source NVR using Node.js and Docker to manage and process 10,000+ concurrent video streams
- Built backend infrastructure enabling secure remote camera access across 50,000 deployed VITEL 4G SIM cards

GistMe Inc

Los Angeles, CA

Software Engineering Intern

June 2024 – August 2024

- Developed a RESTful API using Flask for a web-based trivia platform supporting 2,000+ concurrent users
- Designed scalable data storage using NoSQL databases, reducing average data retrieval latency by 30%

RESEARCH EXPERIENCE

USC QED

Los Angeles, CA

Undergraduate Research Assistant

Jan 2026 – Present

- Analyzed CNN inference latency using the Mobile ML Benchmark (TFLite) dataset across mobile CPU platforms
- Designed feature pipelines for `conv2d` and `fully_connected` layers; evaluated accuracy using Mean Absolute Percent Error
- Conducted hyperparameter tuning with grid search and structured train/validation/test workflows
- Investigating transfer learning approaches for cross-platform latency prediction

PROJECTS

Kestrel | Go

- Implemented the RAFT consensus algorithm to build a fault-tolerant, strongly consistent distributed key-value store
- Designed a linearizable replicated state machine with leader election, log replication, and majority-based commit semantics
- Achieved sub-10ms p99 write latency and 99.9% availability across a 5-node cluster under leader failover
- Built concurrent RPC handlers with mutex-protected shared state to safely handle hundreds of client requests per second

Drift | Go, PostgreSQL, Next.js

- Built a developer tool detecting silent state drift in event-driven systems by replaying event logs and validating canonical state
- Implemented a deterministic Go replay engine capable of processing 100k+ events to detect out-of-order or duplicated events
- Designed an event ingestion pipeline and REST API backed by PostgreSQL supporting low-latency event replay queries
- Created a failure simulation framework injecting distributed anomalies across 19 wallet entities to validate system resilience

Merlin | CUDA C, Python, CuPy

- Implemented CUDA kernels for K-Means and logistic regression achieving up to 32× speedup over NumPy CPU baselines
- Optimized K-Means distance computation using shared-memory tiling reducing global memory accesses by ~70%
- Implemented parallel gradient reduction using a shared-memory tree reduction minimizing `atomicAdd` contention

Peregrine | Go

- Implemented an iterative DNS resolver using UDP sockets and raw DNS wire-format parsing (RFC 1035)
- Built a thread-safe TTL-based caching layer reducing upstream DNS queries by ~60% and improving lookup latency

Mako | OCaml

- Implemented a functional language interpreter with a typed AST, lexer, parser, and evaluation pipeline
- Designed a closure-based execution model supporting lexical scoping, `let rec`, higher-order functions, and tuples

LEADERSHIP

Advanced Robotics Combat

Los Angeles, CA

Project Lead

September 2025 – December 2025

- Led a team of five engineers in the design and construction of a combat robot competing against 50+ teams
- Oversaw subsystem integration, testing, and engineering workflow while managing resource allocation, and competition strategy